

COMP4347 A2 — Individual Contribution Reflection

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Primary Subsystem

I owned the full backend (models, controllers, routes, middleware, tests), the player-side frontend (Quiz, Result, History, Leaderboard, Auth context), and the admin frontend (Dashboard, Question Form, Bulk Import, Protected Routes). As one of three members I shared responsibility for integration, validation, and robustness.

Technical Challenge — Leaderboard Showing Duplicate Users

Initial implementation returned the top 10 *attempts*, so one player could appear multiple times. I replaced the simple query with an aggregation pipeline that groups by user and takes their personal best:

```
// commit 2093816
const entries = await Score.aggregate([
  { $group: { _id: "$userId", username: { $first: "$username" },
    score: { $max: "$score" } } },
  { $sort: { score: -1 } },
  { $limit: 10 },
  { $project: { _id: 0, username: 1, score: 1 } },
]);
```

I also denormalised `username` onto `Score` at submit time so the aggregation needs no `$lookup`. Found via manual testing — seeding one user with multiple attempts exposed duplicate leaderboard entries.

Sequence Diagram — Quiz Flow

See attached diagram below.

Commit History

Hash	Date	What
33b3ee5	2026-04-30	Project scaffold
6812122	2026-05-01	Backend models — User, Question, Score
507f146	2026-05-04	All controllers (auth, quiz, admin)
a851193	2026-05-04	auth + admin middleware
67ad7b1	2026-05-04	Test suite — behaviours specified by me, implementation AI-assisted
8cc56ff	2026-05-01	Auth context + useAuth hook
7964fb3	2026-05-01	Quiz state reducer (LOADING / ACTIVE / SUBMITTED)

Hash	Date	What
d27f928	2026-05-01	Quiz page — question flow, answer locking
318498b	2026-05-01	Result, History, Leaderboard pages
6151870	2026-05-08	Admin dashboard frontend
2093816	2026-05-08	Fix leaderboard — aggregation pipeline, personal best
229928f	2026-05-08	Protected + Admin route wrappers
eb34b03	2026-05-04	Fix Mongoose options validator (Mongoose 8 breaking change)
5996518	2026-05-08	Fix test env — NODE_ENV guard to skip rate limits

Variation — Design Decisions

The image-based questions variation was primarily implemented by Chong An. My contribution was reviewing the integration and verifying the quiz controller's sanitised response included `imageUrl` without exposing `correctIndex`.

Integration & Robustness

Uniform `{ success, data?, error? }` envelope on all endpoints. Validation before every DB write (bulk import validates all items before `insertMany`). Rate limiting on login (10 req/15 min) and submit (5 req/1 min), bypassed in test env. `GET /api/quiz/questions` returns 503 when fewer than 6 active questions exist.

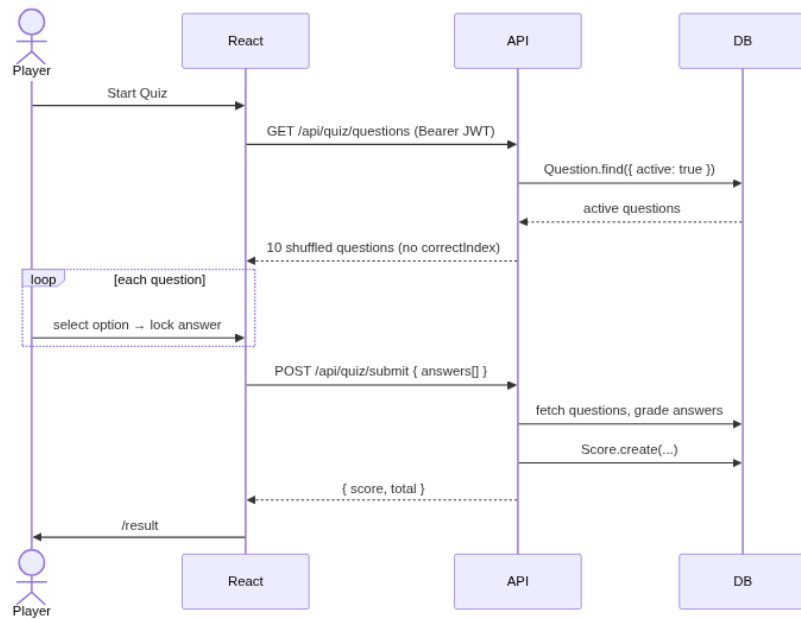


Figure 1: Quiz flow sequence diagram